

MBA
(SEM IV) THEORY EXAMINATION 2022-23
DATABASE MANAGEMENT SYSTEM

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 10 = 20**

- (a) Describe higher impact of failure in DBMS?
- (b) What is Graphical User Interface?
- (c) What is Relation instance?
- (d) Define importance of Notations?
- (e) What is XML?
- (f) Describe SQL Index.
- (g) Describe Isolation in database.
- (h) What is Transaction Log?
- (i) Describe drawbacks of object Database.
- (j) What is data Mining?

SECTION B

2. Attempt any three of the following:**10x3=30**

- (a) Describe Three Schema Architecture of Database Management System?
- (b) Describe fundamental operations of relational algebra are as follows
 - (i) Select (ii) Union (iii) Set different (iv) Cartesian Product
- (c) What do you mean by Data type? Describe Unicode Character Strings Data Types and Binary Data Types?
- (d) What is deadlock? Discuss about the deadlock detection? Also mention the steps for recovery from deadlock?
- (e) Describe Two Tier Client-Server Architecture for Database Management System?

SECTION C

3. Attempt any one part of the following:**10x1=10**

- (a) Construct E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted.
- (b) Describe importance of Menu- Based Interfaces for Web Clients or Browsing in detail?

4. Attempt any one part of the following:**10x1=10**

- (a) What exactly do you mean by Relational Model and its Concepts?
- (b) Describe Domain Relational Calculus using Suitable Example?

5. Attempt any *one* part of the following: 10x1=10

- (a) What are SQL Literals? Explain Integer Literals using suitable example?
- (b) Consider the following Scheme:
SUPPLIER (SUPPLIER ID, SUPPLIER_NAME, SUPPLIER_ADDRESS)
PARTS (PARTS ID, PART_NAME, COLOR)
CATALOG (SUPPLIER ID, PART ID, COST)
Write the following queries in Relational Algebra and in SQL:
 - (i) Find the name of the suppliers who supply Black Parts.
 - (ii) Find the name of suppliers who supply both Blue and Black Parts.
 - (iii) Find the name of suppliers who supply all Parts.

6. Attempt any *one* part of the following: 10x1=10

- (a) Define Serializability using Conflict Serializability?
- (b) What do you understand by concurrency control?

7. Attempt any *one* part of the following: 10x1=10

- (a) What do you understand by decision support Systems?
- (b) Describe features of distributed database?

QP23EP1_128
/ 13-06-2023 13:55:44 | 103.248.120.3